



US 20250280907A1

(19) **United States**

(12) **Patent Application Publication**  
**Agarwal et al.**

(10) **Pub. No.: US 2025/0280907 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **AUXETIC BANDS AND PANELS FOR  
BOTTOM GARMENTS AND GARMENTS  
INCORPORATING THE SAME**

**Publication Classification**

(51) **Int. Cl.**

*A41F 9/02* (2006.01)

*A41D 1/06* (2006.01)

*A41D 1/21* (2018.01)

*A41D 27/20* (2006.01)

*A41D 31/18* (2019.01)

(52) **U.S. Cl.**

CPC ..... *A41F 9/02* (2013.01); *A41D 1/06*

(2013.01); *A41D 1/21* (2018.01); *A41D 27/20*

(2013.01); *A41D 31/18* (2019.02)

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(21) Appl. No.: **19/217,378**

(22) Filed: **May 23, 2025**

**Related U.S. Application Data**

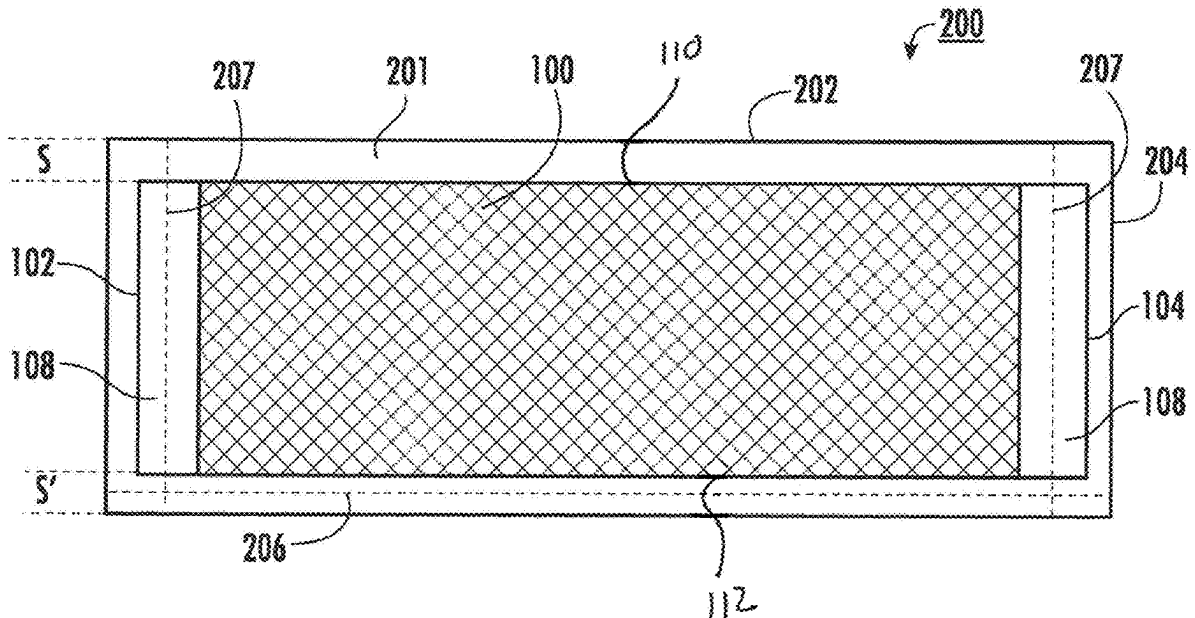
(63) Continuation of application No. PCT/IB2023/  
061752, filed on Nov. 21, 2023.

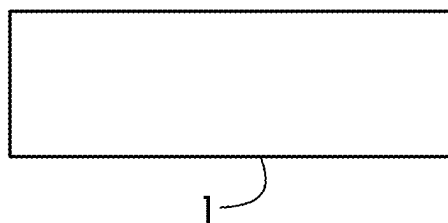
(60) Provisional application No. 63/427,526, filed on Nov.  
23, 2022.

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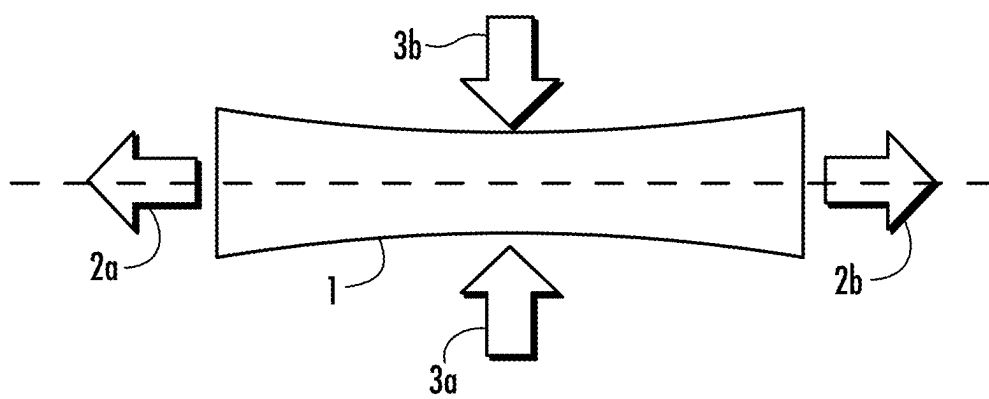
**ABSTRACT**

A garment includes an auxetic device having a first end, a second end opposite the first end, a top edge extending between the first end and the second end, and a bottom edge opposite the top edge. The first end and the second end are fixed to another portion of garment. At least one of the top edge or the bottom edge is free between the first end and the second end. The auxetic device has a Poisson ratio less than or equal to zero when stretched in a direction extending between the first end and the second end.





**FIG. 1**  
PRIOR ART



**FIG. 2**  
PRIOR ART

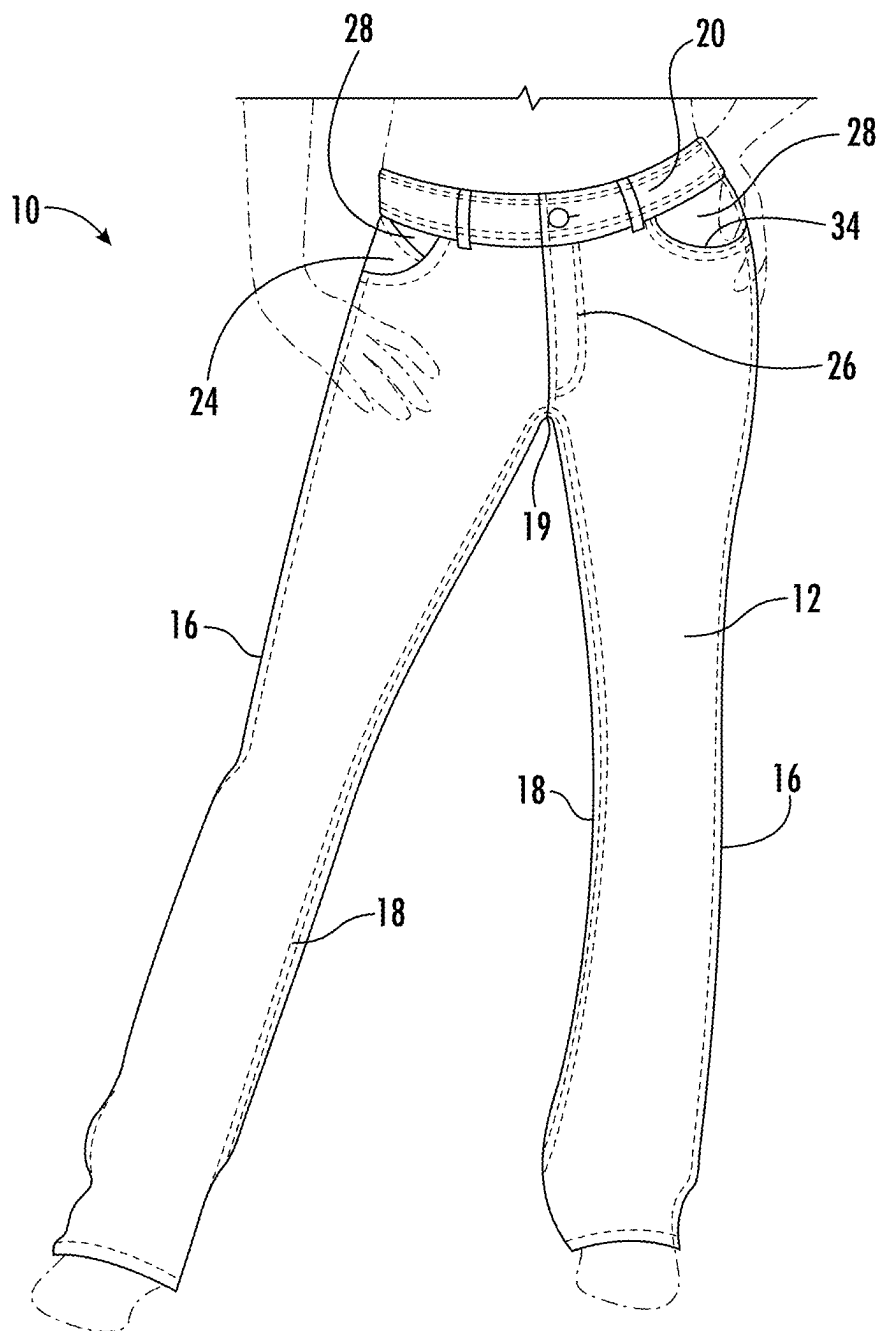


FIG. 3

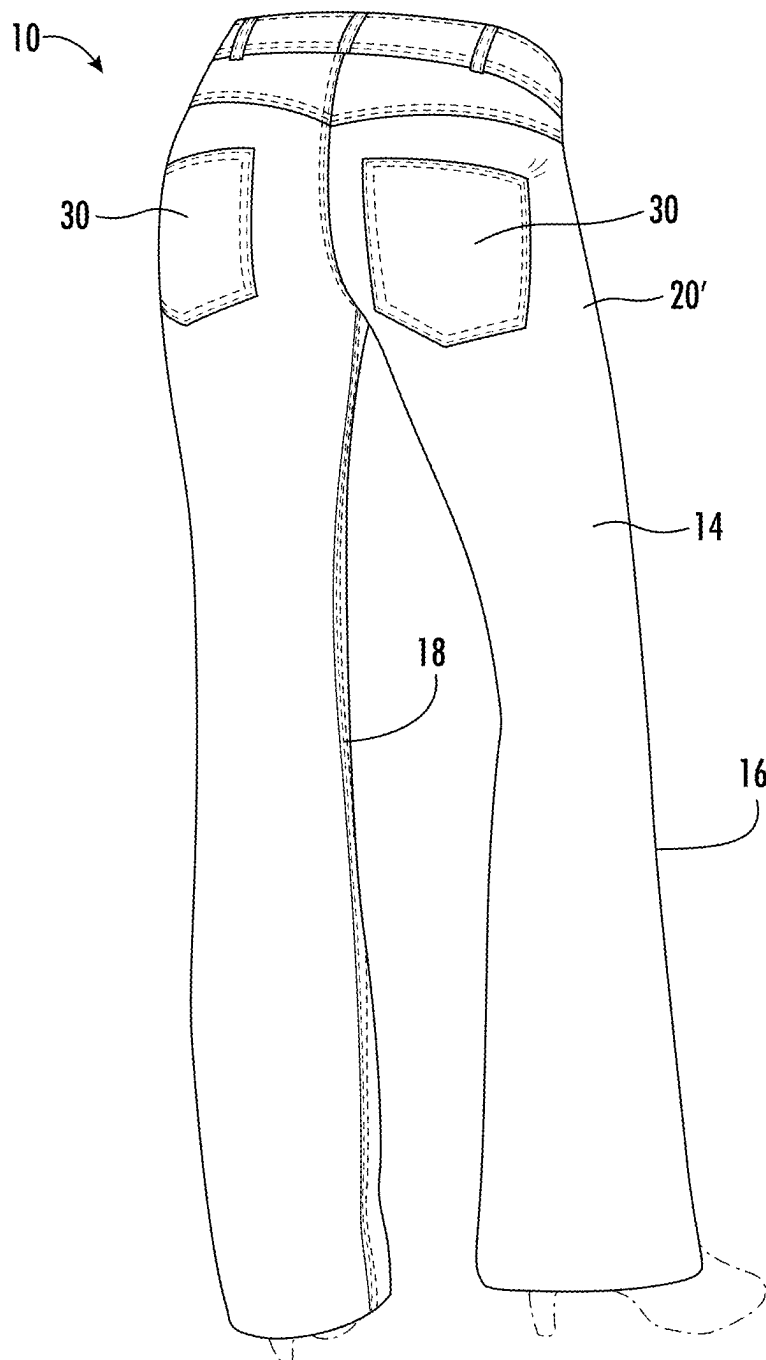
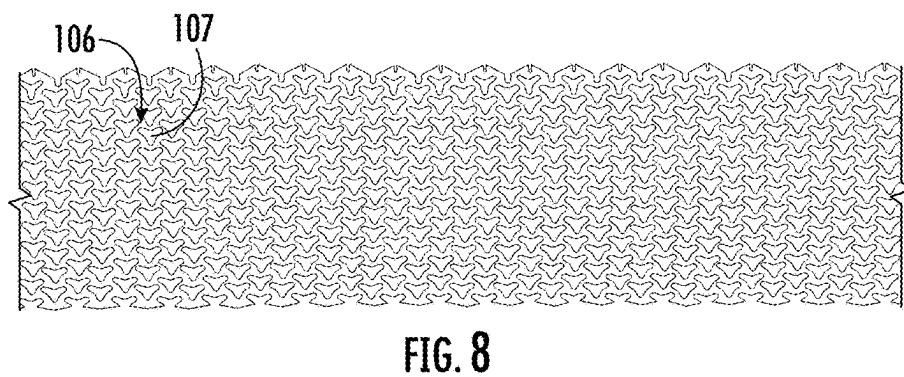
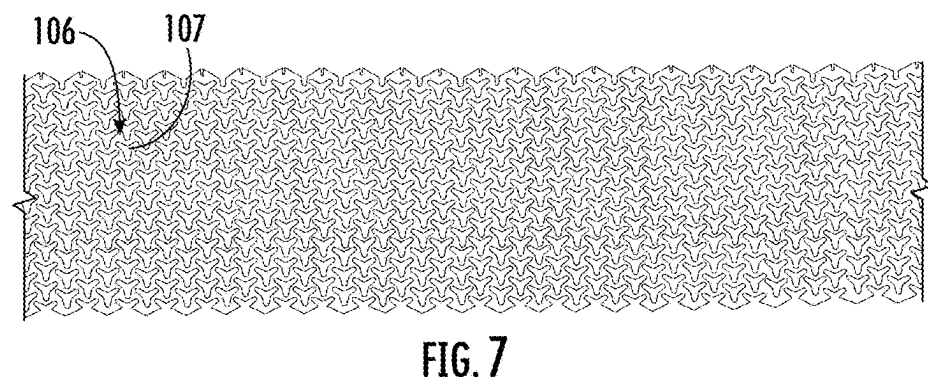
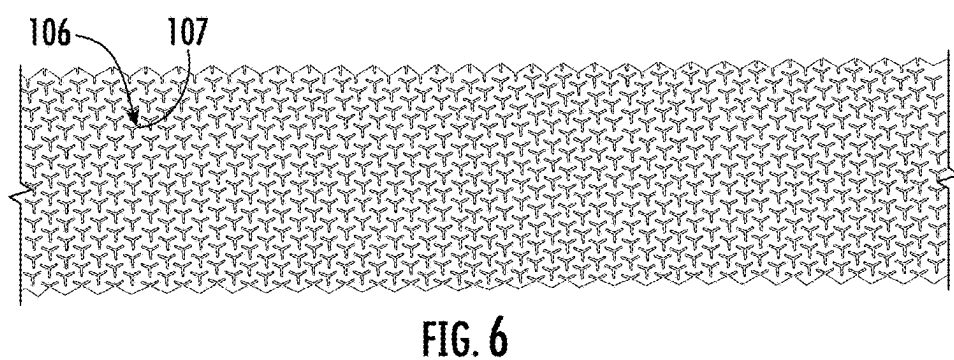
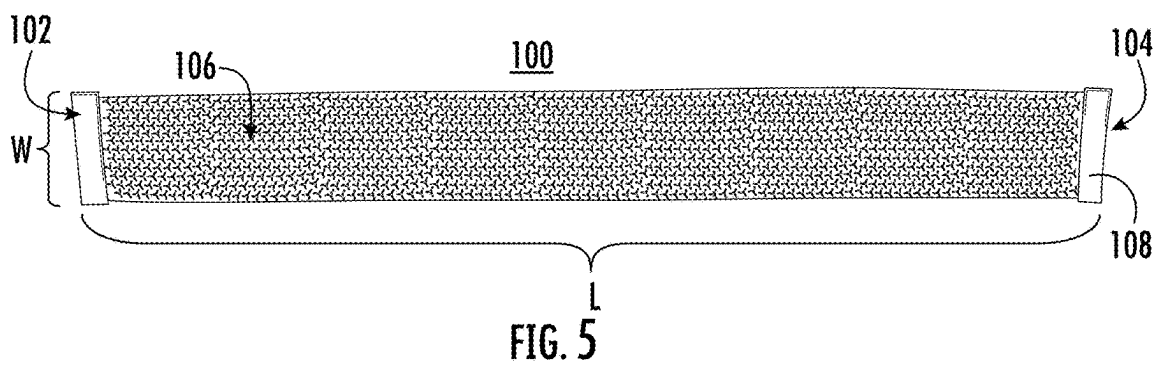


FIG. 4



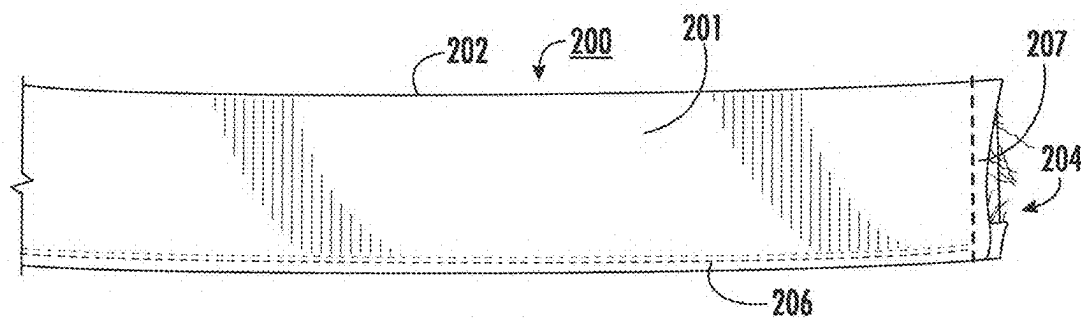


FIG. 9

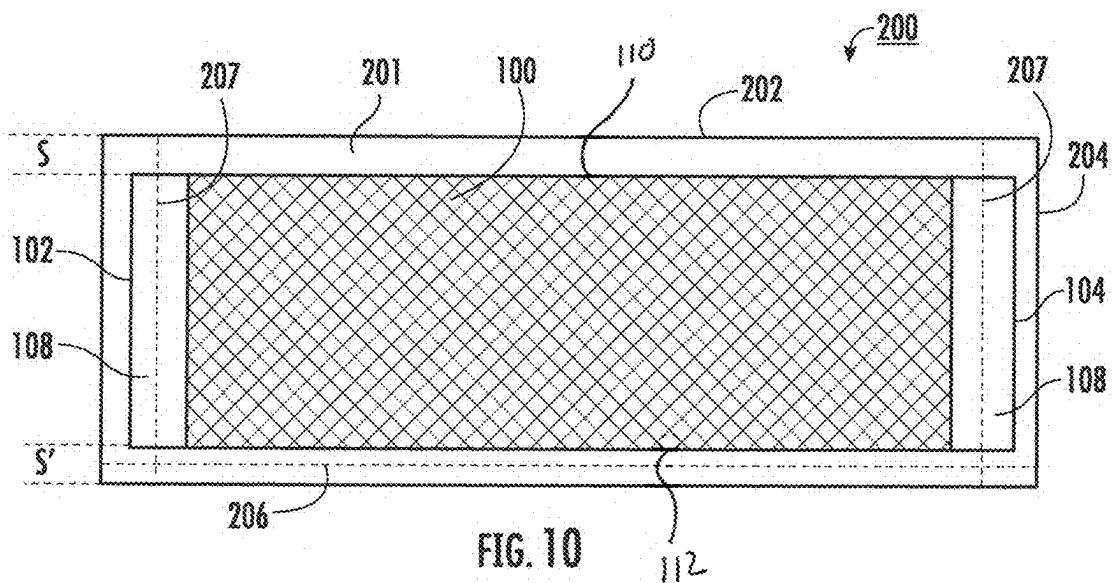


FIG. 10

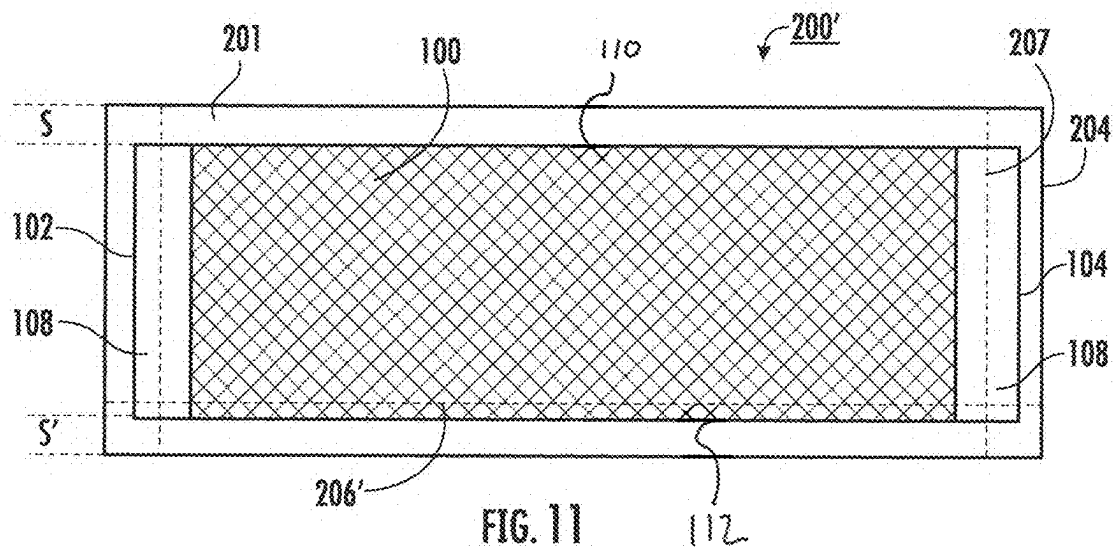


FIG. 11

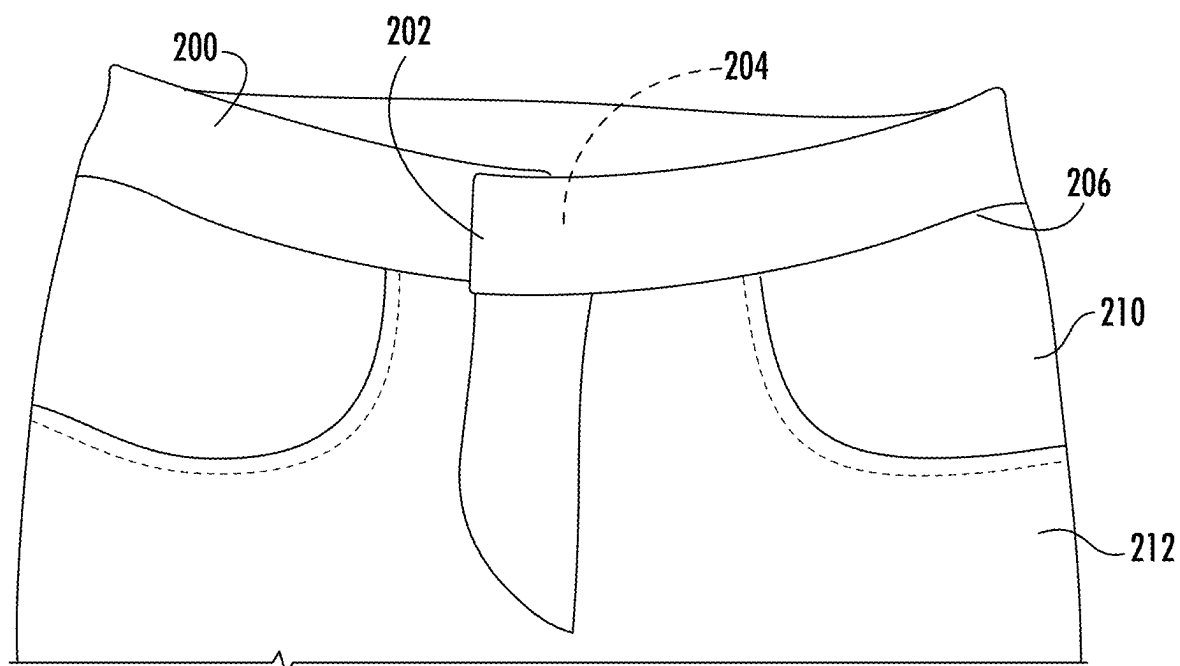


FIG. 12

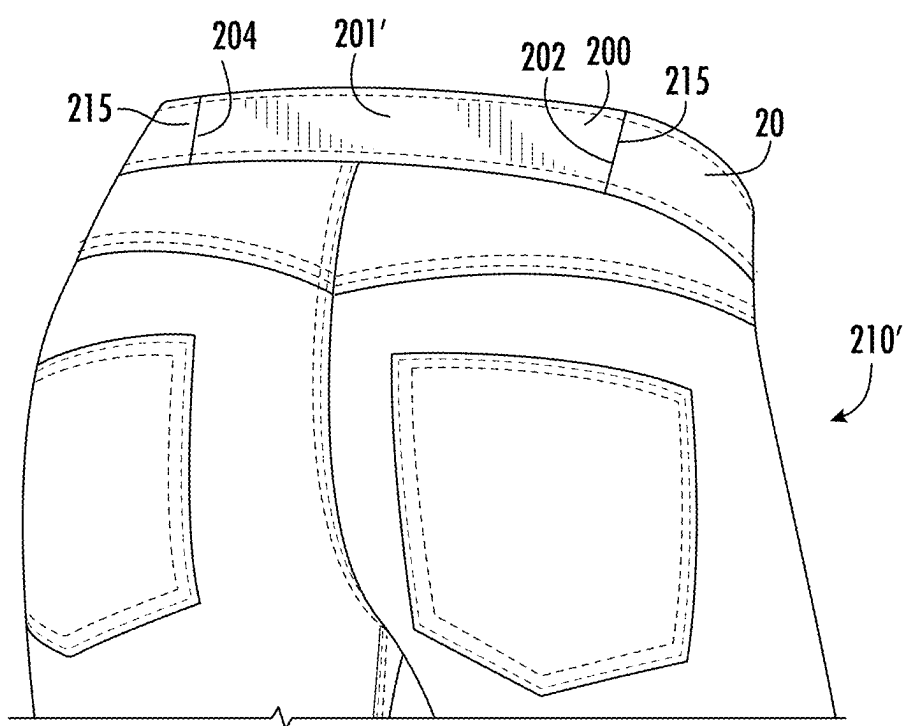


FIG. 13

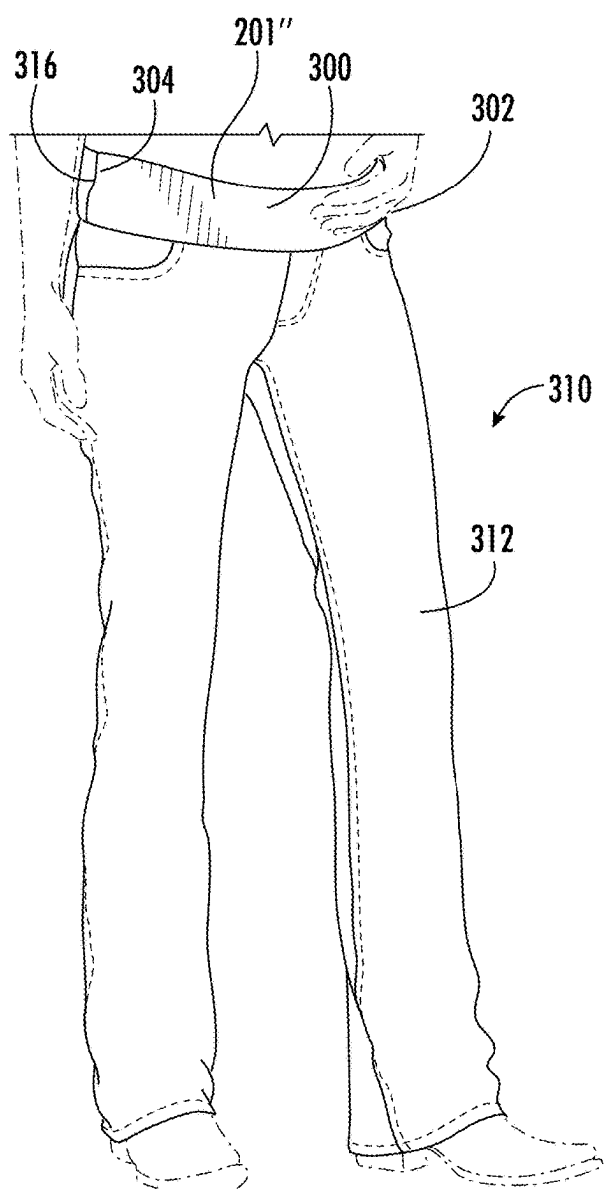


FIG. 14

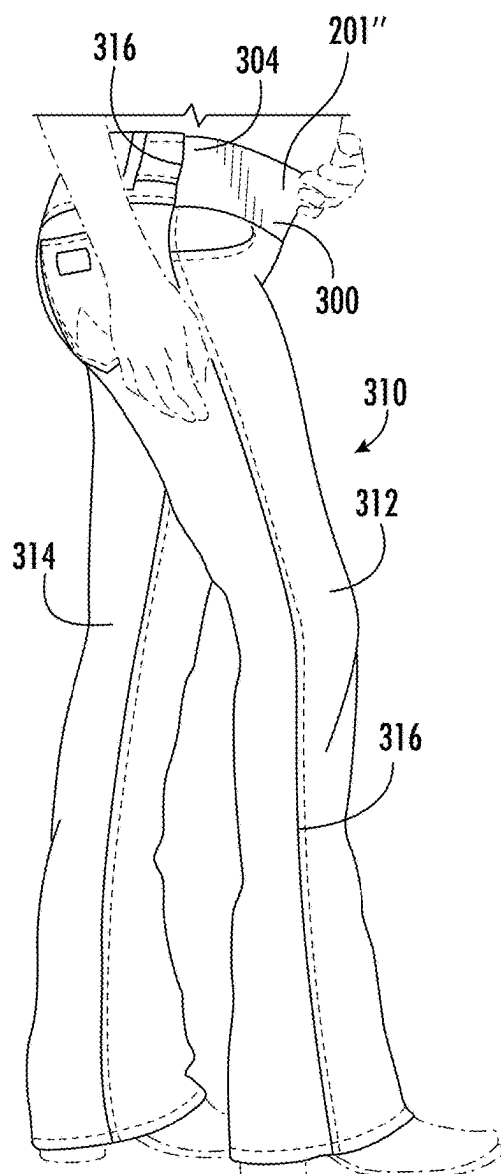


FIG. 15



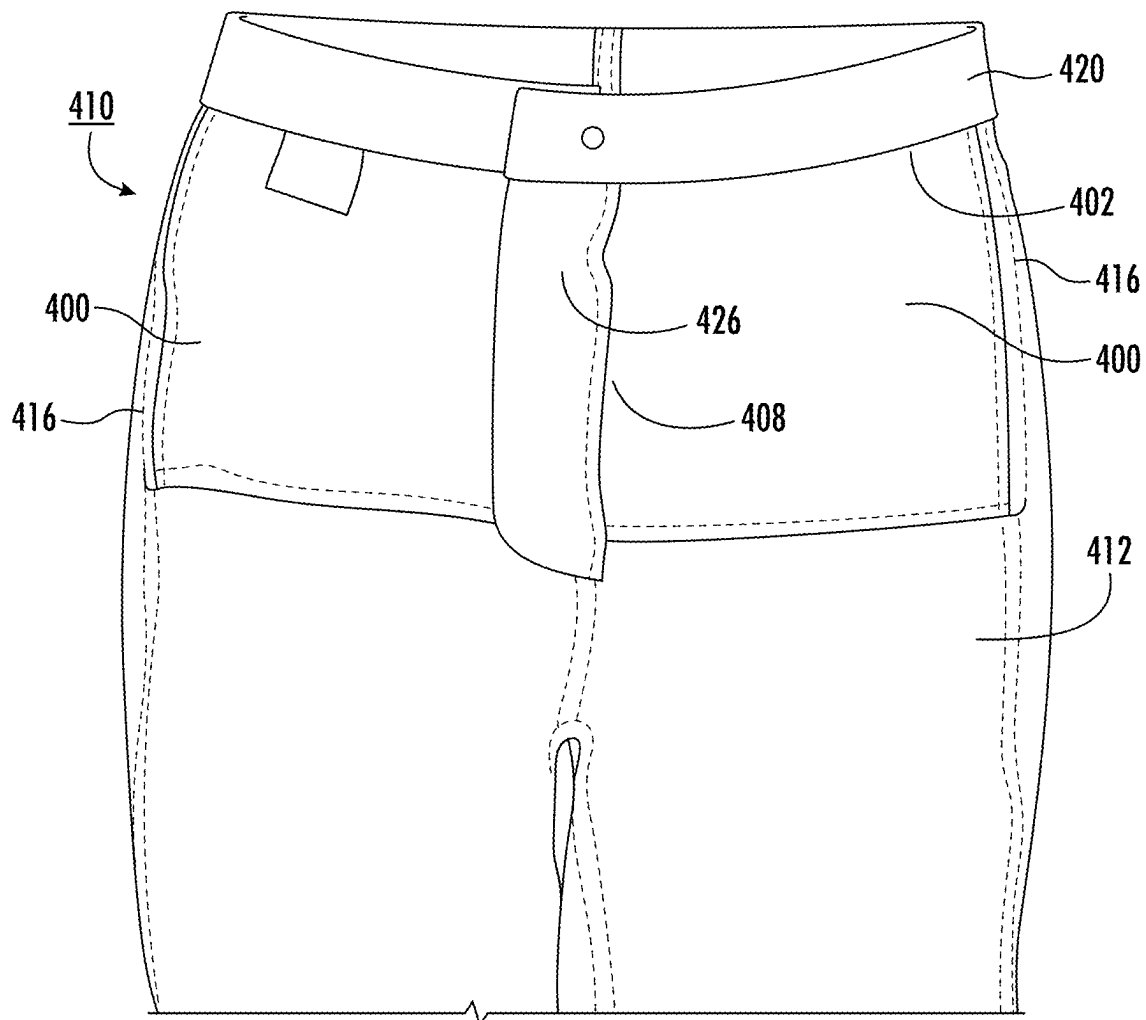


FIG. 16

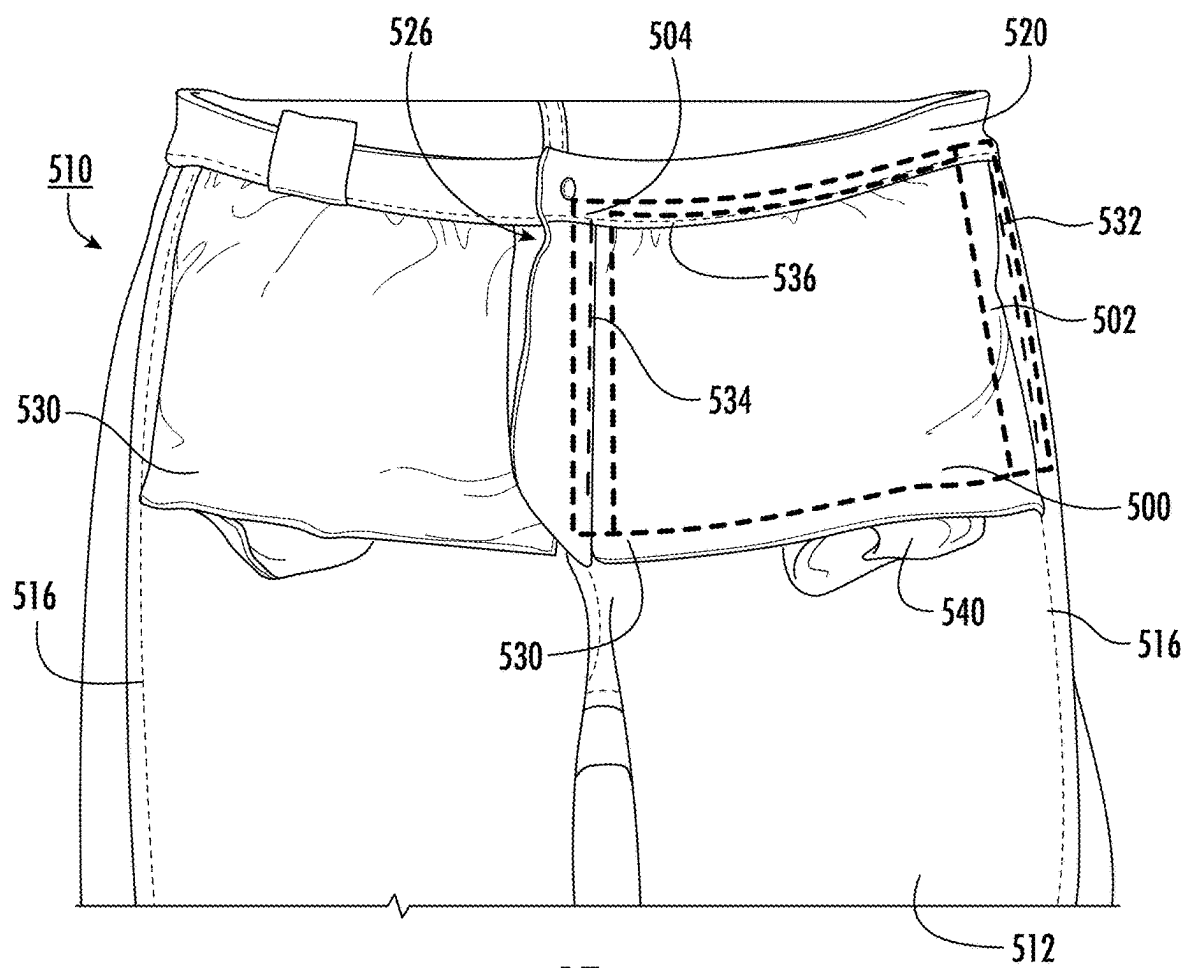


FIG. 17

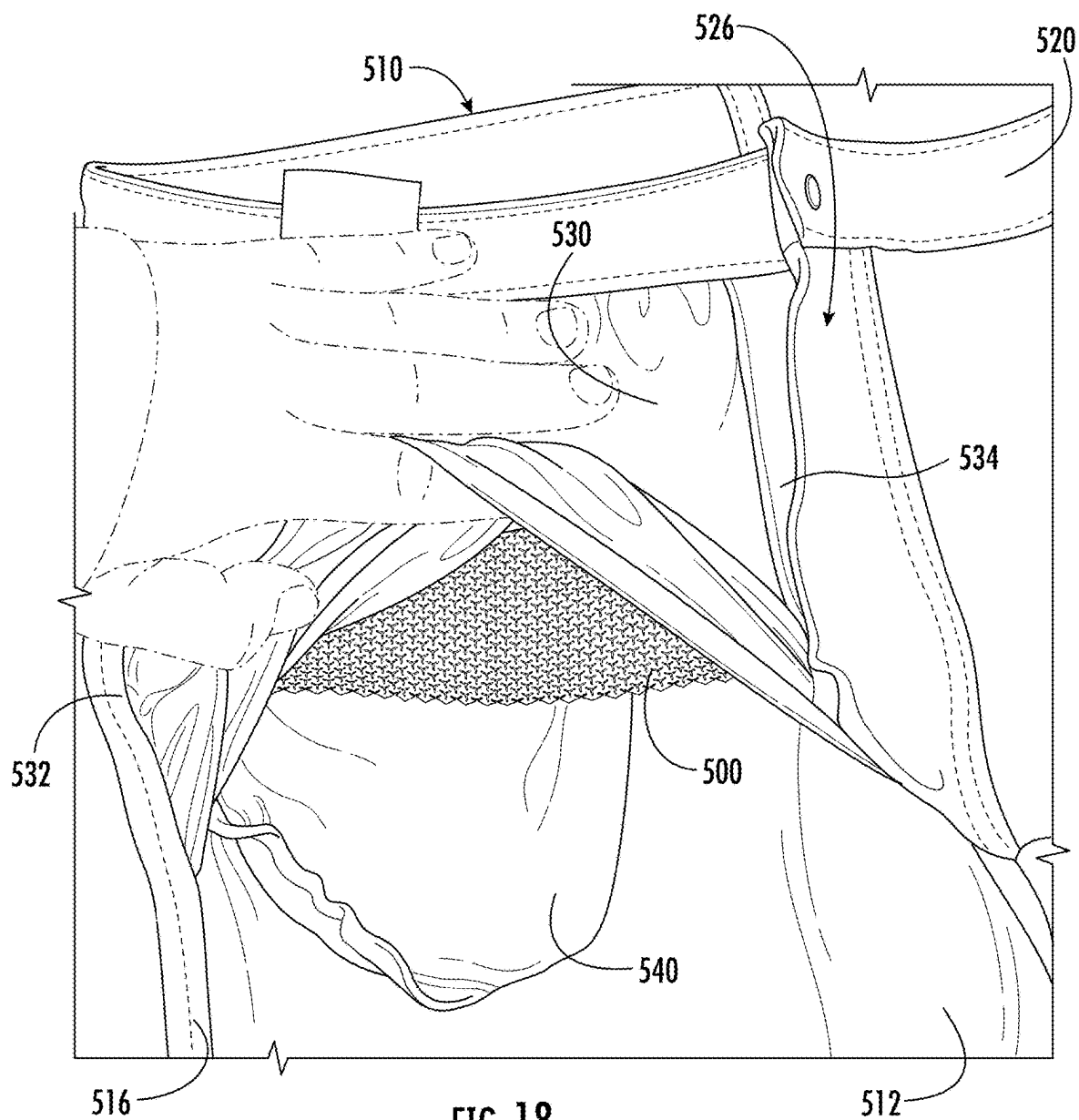


FIG. 18

## AUXETIC BANDS AND PANELS FOR BOTTOM GARMENTS AND GARMENTS INCORPORATING THE SAME

### CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a bypass continuation of WIPO Application Serial No. PCT/IB2023/061752, filed Nov. 21, 2023, which published as WIPO Publication No. WO2024110869, which claims priority to and the benefit of U.S. Provisional Patent Application Ser. No. 63/427,526, filed Nov. 23, 2022. The entire contents of each of the above applications are hereby incorporated by reference.

### BACKGROUND

#### 1. Technical Field

[0002] The present disclosure relates to garments and, more specifically, support bands and panels including auxetic devices for garments.

#### 2. Discussion of Related Art

[0003] Pants and the like (“bottom garments”) are a staple garment found in most closets. Many wearers like the look and practicality of pants but desire a more comfortable fit. Support bands can be incorporated at the top of the bottom garments to help maintain the bottom garment in a desired position on the wearer, usually proximate to the waist of the wearer. In many current commercial examples, support bands are made of an elastic material which, when stretched in one direction (e.g., the x-direction), the other direction (e.g., y-direction) narrows. This is referred to as exhibiting a positive Poisson ratio, i.e., the deformation or narrowing of a material in directions perpendicular to the specific direction of elongation.

[0004] In order to ensure that a bottom garment maintains its position on a body, the elastic support band must have a sufficient restoring force, squeezing the support band against the body of a wearer. However, the higher the force to ensure that the bottom garment does not slide down the body, the greater the stress and potential for discomfort to the wearer. Increasing the resistance of stretching may also result in increased abrasion/chafing of the garment against the skin as well as constriction to movement. In some cases, such as when bending over, current support band stretch and create a “gap situation” wherein the undergarments of the wearer or skin of the wearer may be undesirably exposed.

### SUMMARY

[0005] Various details of the present disclosure are hereinafter summarized to provide a basic understanding. This summary is not an extensive overview of the disclosure and is neither intended to identify certain elements of the disclosure, nor to delineate the scope thereof. Rather, the primary purpose of this summary is to present some concepts of the disclosure in a simplified form prior to the more detailed description that is presented hereinafter.

[0006] Herein described are support band components, support bands, and support panels that utilize an auxetic device to form an auxetic support component, auxetic support band, or auxetic support panel. The auxetic support components, auxetic support bands, and auxetic support panels described herein may be formed of or include auxetic

materials. The auxetic devices detailed herein may be incorporated into or otherwise coupled to a top portion of a bottom garment including, but not limited to, pants, shorts, jeans, leggings, jeggings, capris, skirts, or the like. The auxetic devices detailed herein may be configured to help maintain the garment in a desired position on the body of the wearer. The auxetic devices detailed herein may be incorporated into the interior of a bottom garment, e.g., positioned about a wearer’s belly. The auxetic devices may provide a comfortable and conforming fit to a wearer while keeping the garment in place on the wearer. The auxetic device may provide an increased surface area where stretched which may enhance the comfort to wearer of garment.

[0007] In some embodiments, the auxetic device disclosed herein may be formed of an elastomer material with a negative Poisson ratio such that the auxetic device is configured to expand vertically when it is stretched horizontally. In certain embodiments, the auxetic device disclosed herein may increase in surface area coverage when stretched.

[0008] In an aspect of the present disclosure, a garment includes an auxetic device having a first end, a second end opposite the first end, and a plurality of unit cells disposed between the first end and the second end. The plurality of unit cells have a reentrant structure such that the auxetic device has a Poisson ratio less than or equal to zero when stretched in a direction extending between the first end and the second end.

[0009] In another aspect of the present disclosure, a garment includes an auxetic device having a first end and a second end opposite the first end. The auxetic device has a Poisson ratio of less than or equal to zero when stretched in a direction extending between the first end and the second end.

[0010] In aspects, the auxetic device includes a plurality of unit cells disposed between the first end and the second end. The plurality of unit cells may be arranged to have a reentrant structure.

[0011] In some aspects, the garment further includes a front panel and a rear panel joined together at side seams to form outer edges of the garment. The front panel and the rear panel define a top opening that may be configured to be positioned adjacent a waist of a wearer. The garment may include a support band extending about the top opening. The support band may be configured to maintain a position of the garment relative to the waist of the wearer. The auxetic device may form a portion of the support band. The auxetic device may form a portion of the support band above the rear panel or may form a portion of the support band above the side seams. The first end may be attached to the support band adjacent one of the side seams defined between the front panel and the rear panel and the second end is attached to the support band adjacent the other side of the seam. The front panel includes a fly assembly having a first side and a second side. The fly assembly may be configured to selectively secure one side of the front panel to another side of the front panel. The first end of the auxetic device may be attached to the first side of the fly assembly and the second end may be attached to the second side of the fly assembly. The front panel and the rear panel may be formed of a stretch fabric.

[0012] In certain aspects, the auxetic device includes an edge that extends between the first end and the second end. The edge of the auxetic device may be attached to the front panel or rear panel adjacent the edge.

[0013] In particular aspects, the garment further includes a sleeve that defines a tunnel. The sleeve may be closed about the first end and the second end such that auxetic device is enclosed within the sleeve. The tunnel may be sized and dimensioned to accommodate a stretched state of the auxetic device having an increased surface area such that the auxetic device is capable of expanding in a direction perpendicular to a stretch direction within the tunnel.

[0014] In aspects, the garment further includes a body contact fabric. The auxetic device may be sandwiched between the front panel and the body contact fabric. The body contact fabric may have a first edge and second edge opposite the first edge. The auxetic device may be attached to the front panel or the rear panel along one of the first edge or the second edge. The body contact fabric may be configured to conceal the auxetic device. The first end and the first edge may be stitched together to attach the auxetic device and the body contact fabric to the front panel or the rear panel.

[0015] In another aspect of the present disclosure, an auxetic band for a garment includes an auxetic device and a sleeve of stretch fabric. The auxetic device has a first end and a second end opposite the first end. The auxetic device has a Poisson ratio less than or equal to zero. The sleeve of stretch fabric has a first sleeve end and a second sleeve end opposite the first sleeve end. The sleeve of stretch fabric is dimensioned to house the auxetic device therein. The first end of the auxetic device is fixed to the sleeve adjacent the first sleeve end and the second end of the auxetic device is fixed to the sleeve adjacent the second sleeve end.

[0016] In aspects, the sleeve is dimensioned to accommodate an increase in surface area when the auxetic device is stretched in the direction extending between the first end and the second end.

[0017] In certain aspects, a garment includes an auxetic band configured to maintain a position of the garment relative to a wearer. The garment may include an auxetic band wherein a top edge of the auxetic band may be positioned at or below a waistband of the garment or a bottom edge of the auxetic band may be positioned at a waistband of the garment.

[0018] In another aspect of the present disclosure, a garment includes a support band and an auxetic device. The support band is configured to position a top portion of the garment about or adjacent a waist of a wearer. The auxetic device has a first end and a second end opposite the first end. The auxetic device has a Poisson ratio less than or equal to zero. The auxetic device is secured to the support band.

[0019] In aspects, the auxetic device forms a portion of the support band. The auxetic device may include a top edge secured to the support band or below the support band such that the auxetic device forms a belly support panel of the garment below the support band. The auxetic device may include a bottom edge. The bottom edge may be substantially free from the rest of the garment. The auxetic device may extend above the support band of the garment.

[0020] In some aspects, the garment further includes a front panel and a body contact fabric panel. The body contact fabric panel may be secured to the front panel. The auxetic device may be positioned between the body contact fabric panel and the front panel with the body contact fabric panel preventing the auxetic device from contacting the skin of a wearer.

[0021] In certain aspects, the garment further includes a pocket formed of pocket fabric. The pocket fabric may be disposed between the front panel and the auxetic device.

[0022] Further, to the extent consistent, any of the embodiments or aspects described herein may be used in conjunction with any or all of the other embodiments or aspects described herein.

#### BRIEF DESCRIPTION OF THE DRAWINGS

[0023] Various aspects of the present disclosure are described hereinbelow with reference to the drawings, which are not necessarily drawn to scale, which are incorporated in and constitute a part of this specification, wherein:

[0024] FIG. 1 illustrates a prior art rectangular shaped standard stretch band;

[0025] FIG. 2 illustrates the stretch characteristics of the prior art stretch band of FIG. 1;

[0026] FIG. 3 is a front external view of a bottom garment provided in accordance with an embodiment of the present disclosure;

[0027] FIG. 4 is a back external view of the bottom garment of FIG. 3;

[0028] FIG. 5 is a plan view of an auxetic device provided in accordance with an embodiment of the present disclosure in a neutral or unstretched state;

[0029] FIG. 6 is an enlarged view of a portion of the auxetic device of FIG. 5;

[0030] FIG. 7 is a plan view of the portion of the auxetic device of FIG. 6 in a first stretched state;

[0031] FIG. 8 is a plan view of the portion of the auxetic device of FIG. 6 in a second stretched state;

[0032] FIG. 9 is a plan view of an auxetic band provided in accordance with an embodiment of the present disclosure including a sleeve housing an auxetic device in a neutral or unstretched state;

[0033] FIG. 10 is a cutaway view of the auxetic band of FIG. 9 with an exterior layer of the sleeve removed to expose an interior of a sleeve containing the auxetic device;

[0034] FIG. 11 is the cutaway view of another auxetic band provided in accordance with an embodiment of the present disclosure;

[0035] FIG. 12 is a front exterior view of another garment provided in accordance with an embodiment of the present disclosure incorporating an auxetic band as a support band;

[0036] FIG. 13 is a rear exterior view of another garment provided in accordance with an embodiment of the present disclosure incorporating an auxetic band as a segment of its support band;

[0037] FIG. 14 is an exterior view of another garment provided in accordance with an embodiment of the present disclosure incorporating an auxetic band as a belly panel;

[0038] FIG. 15 is an exterior side view of the garment of FIG. 14;

[0039] FIG. 16 is an interior view of another garment provided in accordance with an embodiment of the present disclosure incorporating auxetic bands as tummy control panels;

[0040] FIG. 17 is an interior view of another garment provided in accordance with an embodiment of the present disclosure incorporating auxetic devices as tummy control panels; and,

[0041] FIG. 18 is an interior view of the garment of FIG. 17 with a body contact fabric of the garment lifted to show the auxetic device.

## DETAILED DESCRIPTION

**[0042]** The present disclosure will now be described more fully hereinafter with reference to example embodiments thereof with reference to the drawings in which like reference numerals designate identical or corresponding elements in each of the several views. These example embodiments are described so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art. Features from one embodiment or aspect can be combined with features from any other embodiment or aspect in any appropriate combination. For example, any individual or collective features of method aspects or embodiments can be applied to apparatus, product, or component aspects or embodiments and vice versa. The disclosure may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. As used in the specification and the appended claims, the singular forms “a,” “an,” “the,” and the like include plural referents unless the context clearly dictates otherwise. In addition, while reference may be made herein to quantitative measures, values, geometric relationships or the like, unless otherwise stated, any one or more if not all of these may be absolute or approximate to account for acceptable variations that may occur, such as those due to manufacturing or engineering tolerances or the like.

**[0043]** The present disclosure relates to articles of manufacture such as support bands, support band sections, support band components, and support panels that utilize auxetic devices. Auxetic devices include structures and/or materials that have a negative Poisson's ratio. Specifically, when an auxetic device is stretched in one direction, e.g., the x-direction, the structure and/or material expands in a direction substantially perpendicular to the applied force, e.g., in the y-direction. When stretched past a threshold in the stretch direction, in a range of 20 percent to 50 percent stretch, the auxetic device may exhibit a positive Poisson ratio. This expansion in the substantially perpendicular direction may occur due the nature and arrangement of the internal structure of the auxetic device. When used as part of a support band or panel, the surface area of the support band or panel increases as the auxetic device is stressed leading to less stress per unit area experienced by a wearer. As used herein, “auxetic” may refer to structure or materials having a negative Poisson's ratio or a Poisson's ratio of zero, i.e., a Poisson's ratio less than or equal to zero, up to a stretch threshold. The stretch threshold may be in a range of 20 percent to 50 percent. In some embodiments, the stretch threshold may be greater than 50 percent.

**[0044]** In typical stretch bands, when the stretch band is stretched in one direction by pulling ends away from each other, the stretch band narrows in a direction perpendicular to the stretch direction. This effect is shown in the prior art elastic band **1** illustrated in FIGS. **1** and **2** in which the stretch direction is indicated by arrows **2a** and **2b** in which the elastic band **1** is stretched or lengthened and in the direction perpendicular to the stretch direction the elastic band **1** narrows as indicated by arrows **3a** and **3b**. This is referred to as the elastic band **1** exhibiting a positive Poisson ratio, i.e., narrowing a material in directions perpendicular to the specific direction of elongation or lengthening.

**[0045]** Referring now to FIGS. **3** and **4**, a garment **10** is provided in accordance with embodiments of the present

disclosure. As shown, the garment **10** is a pair of jeans; however, the garment **10** may be any bottom garment including, but not limited to, pants, shorts, jeans, leggings, jeggings, capris, skirts, or the like. In some embodiments, the garment **10** may be a top garment including, but not limited to, a shirt, blouse, t-shirt, sweater, sports bra, bralet, brazier, halter top, or the like. In certain embodiments, the garment **10** may be another type of garment including, but not limited to, a dress, a jump suit, an under garment, a sock, or the like. FIG. **3** illustrates a front view of the garment **10** as worn by a wearer. The garment **10** may include several sub-assemblies such as a front panel **12**, a back panel **14**, and a support band **20** that are coupled together. For example, the front panel **12** and the back panel **14** may be coupled together with stitching that form side seams **16** and inseams **18**. The side seams **16** extend from a bottom of the garment **10** to the support band **20** adjacent the top of the garment **10**. The side seams **16** may form the side edges of the garment **10**. The inseams **18** extend from the bottom of the garment **10** to a crotch point **19**. The garment **10** may include pocket assemblies **24** that may each include a pocket facing **28**.

**[0046]** The garment **10** may include a fly fastener assembly **26**. The fly fastener assembly **26** may fasten opposite sides of the front panel **12** together with a variety of fasteners including, but not limited to, zippers, buttons, hook and loop fasteners, hook and eye snap, string ties, or other conventional fly assemblies. The fly assembly **26** may include a first fly portion **27** and a second fly portion **29** (not illustrated) that are each coupled to the front panel **12** and/or the support band **20**. The first fly portion **27** and/or the second fly portion **29** may be coupled to the front panel **12** by stitching.

**[0047]** The support band **20** may be referred to as a waistband. The support band **20** is generally positioned at the top of the garment and may extend around a perimeter of the garment **10** from the front of the bottom garment **10** around the back of bottom garment **10**.

**[0048]** Turning to FIG. **4** which illustrates a back view of the garment **10** worn by a wearer. The back panel **14** may include one or more pocket assemblies **30**. The pocket assemblies **30** may be a patch pocket, welt pocket, insert pocket, hidden pockets, or other conventional pockets.

**[0049]** The front and back panels **12**, **14** may be formed of a denim fabric material. In some embodiments, the denim material may be a high weft-stretch fabric. The front and back panels **12**, **14** may, for example, be a stretch denim fabric for jeans and/or other fabrics used for other bottom garments. In examples, the stretch of the front panel **12** or the back panel **14** in the weft direction is greater than 25 percent when tested according to ASTM D2594-04 (2012). In some examples, the stretch of the front panel **12** or the back panel **14** in the weft direction may be in a range of 25 percent and 60 percent when tested according to ASTM D2594-04 (2012). In embodiments, the stretch of the front panel **12** and the back panel **14** in the weft direction is 50 percent when tested according to ASTM D2594-04 (2012). In some embodiments, the front and back panels **12** and **14** may have a stretch exceeding 50 percent in the weft direction when tested according to ASTM D2594-04 (2012). In certain embodiments, the stretch of the front and back panels **12**, **14** may be less than 25 percent when tested according to ASTM D2594-04 (2012). It is to be appreciated that while the present disclosure is discussed in relation to denim fabric material for the garment **10**, other fabrics known in the art

for garments, including but not limited to, khaki, corduroy, linen, polyester, cotton twill, hemp, or the like may be substituted for one or more of the assemblies of the garment 10.

[0050] With reference to FIG. 5, an auxetic device 100 is provided in accordance with embodiments of the present disclosure. The auxetic device 100 may enhance the comfort of the garment 10 for a wearer. The auxetic device 100 is generally of an elongated construction having a length L, extending from a first end 102 to a second end 104. In such an elongated construction, the length L is greater than the width W of the auxetic device 100. In some embodiments, the width W ranges from 0.5 inches to 10 inches. In embodiments where the auxetic device 100 is part of a support band 20, as detailed below, the width W may be in a range of 1 inch to 2 inches, e.g., 1.75 inches. In embodiments where the auxetic device 100 is part of a tummy panel, discussed in greater detail below, the width W may be in a range of 2 inches to 9 inches. The length L of the auxetic device 100 may vary per desired application as well as garment size. For example, a garment with a 30-inch waist dimension (the approximate perimeter of the interior opening of the garment about the top of the garment when the fly fastener assembly 26 is engaged) may have a shorter length L than a garment with a 38-inch waist dimension. The length L may be configured to wrap entirely around the waist of a wearer. The length L may be configured to partially wrap around the waist of the wearer. In some embodiments, multiple auxetic devices 100 may be incorporated into a single garment. For example, multiple auxetic devices 100 may be incorporated into a single support band 20. In such embodiments, the auxetic devices 100 may be spaced apart from one another. In some embodiments, the auxetic devices 100 may be disposed above the side seams with spaces or gaps between the auxetic devices 100 above the front panel and the back panel. In certain embodiments, the auxetic devices 100 may be disposed above the front panel and the back panel with spaces or gaps between the auxetic devices 100 above the side seams.

[0051] The auxetic device 100 has a thickness in a range of 1.5 mm to 4 mm, e.g., 2 mm to 2.2 mm. In some embodiments the thickness of the auxetic device 100 may be less than 1.5 mm or greater than 4 mm. The thickness of the auxetic device 100 may be selected based on the application of the auxetic device 100. For example, the thickness T may be greater when the auxetic device 100 will be subject to high forces to prevent tearing or breaking of the cell structure of the auxetic device 100. In addition, when less force is required from the auxetic device, the thickness T may be reduced to aid in incorporation into a garment using conventional tools, e.g., sewing needles, or to avoid the auxetic device 100 be considered as bulky or uncomfortable to a wearer.

[0052] Continuing to refer to FIGS. 5-8, the auxetic device 100 has a first end 102 and a second end 104 with a plurality of unit cells 106 defined therebetween. The plurality of unit cells 106 are arranged to provide auxetic properties. The unit cells 106 may be arranged in an array extending between the first end 102 and the second end 104. In some embodiments and as illustrated in FIG. 4, the first end 102 and/or the second end 104 may include a solid area portion 108 that does not contain unit cells 106 and/or voids 107. The solid area portion 108 may be used to facilitate connection of the auxetic device 100 to a garment 10 and/or sleeve 201 as

described in detail below. For example and without limitation, the solid area portions 108 may be configured to receive stitching to couple the solid area portions 108 at a desired position of a garment or a retention sleeve.

[0053] Each of the unit cells 106 disposed between the first end 102 and the second end 104 include a structure (e.g., reentrant structure) having a size, shape, and orientation such that when the auxetic device 100 is stretched along its length L, the auxetic device 100 has a Poisson's ratio equal to or less than zero, i.e., maintains its width W or expands along its width W. In some examples, the width W of the auxetic device 100 increases from 0 percent to 50 percent, e.g., 5 percent to 25 percent, as the auxetic device 100 is stretched along its length L. In some embodiments, the width increases by 15 percent as the auxetic device 100 is stretched along its length L. In an embodiment when the auxetic device 100 is disposed within a support band 20, the width W may expand from 1.75 inches to 2 inches when stretched. An example of width W expansion when the auxetic device 100 is stretched along the length L is shown in FIGS. 6-8.

[0054] Each cell unit 106 includes a void 107 that increases in a void area as the auxetic device 100 is stretched along its length L and maintains or expands in its width W. With particular reference to FIG. 6, the void 107 of a cell unit 106 has minimal area before being stretched such that the cell unit 106 is in a relaxed or unstretched state. FIG. 7 illustrates the void 107 of a cell unit 106 when the auxetic device 100 is in a first stretched state such that the void 107 has an increased void area while the opposing ends of the auxetic device 100 are stretched away from each other a first distance or percentage of stretch. The auxetic device 100 may continue to be stretched beyond the first stretched state to a second stretched state as shown in FIG. 8. In the second stretched state the void 107 of a unit cell 106 may have a substantially maximum void area. It is to be appreciated that while certain stretched/expanded states are illustrated, the auxetic device 100 is capable of many stretched/expanded states between the relaxed state and the second stretched state. Furthermore, the honeycomb arrangement of unit cells 106 and the substantially triangular void shape 107 illustrated in FIGS. 5-8 are representative only and that the arrangement, relative size, and reentrant structure shapes may be modified for a particular application such that other reentrant structures and combination of multiple structures may be used for the auxetic devices 100 described herein. It is further noted that as the auxetic device 100 is stretched from the relaxed state toward the second stretched state, the auxetic device 100 increases in relative surface area that the auxetic device 100 covers. It is noted that this increase in relative surface area is a function of both the increase in length and the maintenance or increase in width of the auxetic device 100.

[0055] With reference to FIGS. 9-11, an auxetic band 200 is provided in accordance with the present disclosure. The auxetic band 200 includes a sleeve or casing 201 that encloses or houses the auxetic device 100 therewithin. The sleeve 201 is configured to have substantially the same shape as the auxetic device 100 such that the sleeve 201 is of similar length and similar width. Construction of the sleeve 201 may vary to surround the auxetic device 100 on all sides while allowing for stretching and expansion of the auxetic device 100. As shown in FIG. 10, the auxetic device 100 is placed on an interior-facing surface of fabric forming

the sleeve **201** with the fabric folded over the auxetic device **100** about a fold edge **202**. The fabric forming the sleeve **201** is secured to itself with bottom stitching **206** to enclose the auxetic device **100** within an interior volume or tunnel of the sleeve **201**. The bottom stitching **206** is used to keep the folded arrangement of the sleeve **201**. As shown, the fabric forming the sleeve **201** is secured by stitching; however, in some embodiments, the fabric forming the sleeve **201** is secured by button or snap fasteners, stitching, hook and loop systems, adhesive, or combinations thereof. As shown in FIG. 9, in some embodiments, other configurations of sleeves for housing an auxetic device **100** within the tunnel are considered. For example, a continuous sleeve of fabric may be constricted via known methods, e.g., knitting like a sock, such that there may not be a need to fold the fabric about itself and securing the fabric to itself.

[0056] Each end **204** of the sleeve **201** may be closed to enclose the auxetic device **100** within the tunnel of the sleeve **201**. In some embodiments, the closure about each end **204** is made via stitching **207**. In certain embodiments, the stitching is configured to fix the ends **102**, **104** of the auxetic device **100** to the sleeve **201**. In particular embodiments, the stitching **207** extends through the fixed area **108** of each end **102**, **104**. The construction of the sleeve **201** detailed herein should not be seen as limiting and other constructions of the sleeve **201** are contemplated. For example and without limitation, rather than folding over a single piece of fabric, an auxetic device **100** may be sandwiched between two pieces of fabric and then closed on at least one side by the two pieces of fabric being secured together to creating an auxetic band **200**, e.g., via stitching, button or snap fasteners, hook and loop systems, adhesive, or combinations thereof.

[0057] Continuing to refer to FIG. 10, the auxetic band **200** includes an auxetic device **100** that is encased within the tunnel of the sleeve **201**. The auxetic device **100** is fixed within the tunnel of the sleeve **200** by coupling the ends **102**, **104** to the sleeve **201** such that when the sleeve **201** is stretched, the auxetic device **100** also stretches. A top edge **110** and a bottom edge **112** of the auxetic **100** may be free to move within the sleeve **201**. In some embodiments, the top edge **110** or the bottom edge **112** may be attached, e.g., stitched, to the sleeve **201**. As detailed above, when the auxetic device **100** stretches, the auxetic device **100** may expand. The sleeve **200** may be dimensioned slightly larger than the unstretched width  $W$  of the auxetic device **100**, to allow room in the tunnel of the sleeve **201** to accommodate the expansion of the auxetic device **100** when the sleeve **201** and connected auxetic device **100** are stretched along the length. In certain embodiments, the material of the sleeve **200** may contract in a direction perpendicular to the stretch direction such that the additional width of the sleeve **200** may prevent the sleeve **200** from inhibiting the expansion of the auxetic device **100**. For example, the auxetic device **100** may be fixed within the sleeve **201** such that there is a distance  $S$  between the top of the auxetic device **100** and top edge of the sleeve **202** and a distance  $S'$  between the bottom of the auxetic device **100** and bottom edge of the sleeve **202**. In some embodiments, the distances  $S$  and  $S'$  are identical. In embodiments, the distance  $S$  is different from the distance  $S'$ . The distances  $S$  and  $S'$  may be determined based on the desired expansion of the auxetic device **100** in the width  $W$  direction and/or the contraction of the sleeve **200**. The distances  $S$ ,  $S'$  may be configured to limit or make expansion

more difficult at a certain stretched distance. In some embodiments, the distance  $S$  or  $S'$  may be 0.125 inches when the sleeve **200** forms a waistband of a garment.

[0058] The spacing distances  $S$ ,  $S'$  may prevent deformation of the auxetic band **100** when the sleeve material is a 2-way stretch fabric. The 2-way stretch fabric of the sleeve **200** could restrict expansion of the auxetic band **100** in the width  $W$  direction when the sleeve **200** the spacing distances  $S$ ,  $S'$  do not provide sufficient room for expansion of the auxetic band **100** and/or contraction of the sleeve **200**. In such embodiments, the auxetic device **100** may deform three dimensionally, e.g., create wavy areas, which may be uncomfortable to the wearer when the auxetic band **100** is restricted by the sleeve **200**. In embodiments where the sleeve material is a 4-way stretch fabric, the distances  $S$ ,  $S'$  may be reduced when compared to the distances  $S$ ,  $S'$  for a 2-way stretch fabric. The reduction in the distances  $S$ ,  $S'$  may be possible as a result of the sleeve material stretching or expanding in the direction of expansion of the auxetic device **100** when contacted by the auxetic device **100**.

[0059] Referring now to FIG. 11, another auxetic band **200'** is provided in accordance with the present disclosure. The auxetic band **200'** is similar to the auxetic band **200** with similar labels representing similar components which will not be detailed herein except for the differences with the auxetic band **200**. The auxetic band **200'** includes bottom stitching **206'** that fixes the bottom edge of the auxetic device **100** relative to the sleeve **201**. In this way, the expansion of the auxetic device **100** is limited in the top portion of the auxetic band **200'**. As shown, the entire bottom edge of the auxetic device **100** is fixed to the sleeve **201**; however, in some embodiments, only portions of the bottom edge of the auxetic device **100** are fixed to the sleeve **201**. Fixing the bottom edge or portions of the bottom edge to the sleeve **201** may minimize potential twisting of the auxetic device **100** within the sleeve **201**.

[0060] With reference to FIG. 12, a garment **210** is provided in accordance with the embodiment of the present disclosure. The garment **210** includes the auxetic band **200** that encircles the entire waistline of the garment **210**. As shown, the auxetic band **200** functions as a support band of the garment **210**. The auxetic band **200** may grip a waistline of a wearer to maintain a position of the garment **210** relative to the waistline of the wearer. The auxetic band **200** includes an auxetic device (not explicitly shown) encased in a sleeve **201**. The auxetic band **200** may be secured to the front and/or rear panels of the garment **210** by stitching. In some embodiments, the bottom stitching **206** may be used to secure the auxetic band **200** to the front and/or rear panels of the garment **210**. In some embodiments, the auxetic band **200** may have each of its ends **202**, **204** secured together such that the auxetic band **200** forms a loop around the top of the garment **210**. In certain embodiments, the ends **202**, **204** of the auxetic band **200** may terminate adjacent one another. For example, the ends **202**, **204** of the auxetic band **200** may terminate on opposite sides of a fly area of the garment **210** such that the support band can be fastened at the fly area to functionally close the loop of the auxetic band **200**.

[0061] In embodiments and with reference to FIG. 13, the auxetic band **200** may be incorporated as a discrete section of a support band **20** of a garment **210'** provided in accordance with the present disclosure. The auxetic band **200** includes an auxetic device (not explicitly shown) encased



within a sleeve 201'. As shown, the auxetic band 200 forms a portion of the support band 20 that extends above the rear panel of the garment 210'. The ends of the auxetic band 202, 204 may connect to portions of the support band 20 at connection points 215. The support band 20 may be a stretch type fabric, e.g., a two-way stretch fabric. As shown, the auxetic band 200 extends only a portion of an extent of the support band 20 above the rear panel; however, the auxetic band 200 may extend past one or more of the side seams of the garment 210' or around the entire top of the garment 210'. Furthermore, while illustrated as being centrally positioned on the rear of the garment 210', the auxetic band 200, or multiple auxetic bands, may be positioned anywhere along the support band 20. For example and without limitation, a length of auxetic band 200 may be placed on each side of the garment 210', such as above the pockets of the garment 210' or above each side seam of the garment 210'.

[0062] In some embodiments, an auxetic device, similar to auxetic device 100, is directly used as a portion or portions of the support band 20 instead of disposing the auxetic device within a sleeve such as sleeve 201'. In such embodiments, the ends of the auxetic device may be secured to ends of the support band, e.g., at attachment points 215. In some embodiments the bottom edge of such the auxetic device may be secured to the top edge of the rear panel of the garment. In particular embodiments, a portion of the support band 20 extends between the auxetic device 100 and the wearer to prevent the auxetic device 100 from contacting the skin of the wearer.

[0063] In the embodiment of FIG. 13, placing the auxetic band 200 above the middle of the rear panel may prevent or reduce a "gap situation" in which undergarments of a wearer or skin of a wearer are undesirably exposed when the wearer bends over. For example, when a wearer bends over such that a horizontal stretch along the support band or waistband 20 of the garment 210' is induced, the auxetic band 200 expands vertically in response to the horizontal stretch of the waistband 20. The vertical expansion of the auxetic band may prevent exposure of skin or undergarments when a support band of a garment is stretched.

[0064] FIGS. 14 and 15 illustrate an auxetic band sized and configured as an auxetic belly panel 300 provided in accordance with the present disclosure. As shown, an auxetic device (not explicitly shown) is encased in a sleeve 201" such that the auxetic band is in the form of the auxetic belly panel 300. In this embodiment, the auxetic belly panel 300 is attached to the front panel 312 of the garment 310. For example and without limitation, a bottom portion of the auxetic panel 300 may be sewn or otherwise stitched to a top portion of the front panel 312. In some embodiments, the front panel 312 may be shortened to allow for a belly panel 300 with an extended height. In certain embodiments, a typical closure mechanism, such as a button or fly fastener incorporated into a front panel, is replaced with the auxetic belly panel 300 which provides support to the wearer and helps maintain the position of the bottom garment 310 on the user. In some embodiments, the auxetic belly panel 300 may comfortably conform or constrict portions of the wearer.

[0065] Each end 302, 304 of the auxetic belly panel 300 may attach to the garment 310 above side seams 316 of the bottom garment 310. For example, the ends 302, 304 may be stitched to the rear panel 314 and the ends 302, 304 may be positioned in line with the side seam 316. In embodiments, within the auxetic belly panel 300, the bottom edge 112 of

the auxetic device may be attached to the top portion of the front panel 312 and the top edge 110 may be free within the sleeve 201". The auxetic belly panel 300 may be used in maternity pants to provide more comfort for pregnant wearers. For example, increasing a surface area as the auxetic belly panel 300 is stretched may reduce discomfort to a wearer. That is, the pressure (force/area) experienced by the wearer is reduced by increased surface area of the panel. This is in contrast to typical elastic panels which may narrow and/or decrease in surface area as the material stretches which may cause discomfort, such as an increase in experienced pressure to the wearer as the material stretches. The increase in surface area may relieve the pregnant individual of compressive stress in the belly area while also maintaining the garment 310 in proper position on the body of the wearer. The dimensions of the auxetic belly panel 300 may be larger than the auxetic band configurations described with respect to the above waistband applications. For example, the auxetic belly panel 300 may have a width in a range of 4 inches to 10 inches in the non-expanded or relaxed state. In some embodiments the auxetic belly panel 300 may be positioned above different portions of a garment. For example, the auxetic belly panel 300 may be positioned above the side seams, the rear panel, or may surround the entire circumference of the top of the garment. In certain embodiments, the auxetic belly panel 300 is provided as separate portions that are disposed about the top circumference of the garment.

[0066] In some embodiments, the auxetic device 100 is directly used as the belly panel 20. That is, there is no intermediating sleeve 201 for housing the auxetic device 100. Rather, a bottom portion of the auxetic device 100 may be sewn or otherwise stitched to a top portion of a shortened front panel 312 and ends 104 and 108 are secured along the side seams 316. In some embodiments, an intermediate layer, such as a piece of fabric, may be positioned between the auxetic device 100 and the wearer, avoiding direct contact of the auxetic device with the skin of the wearer.

[0067] FIG. 16 illustrates an inside out view of a garment 410 provided in accordance with the present disclosure. The garment 410 has an auxetic panel 400 which is similar to the auxetic band 200 with like elements having similar labels with a preceding "4" replacing the preceding "2" of the similar element with only the differences detailed herein for brevity. The auxetic panel 400 may be sized and configured as an interior auxetic tummy control panel 400. In FIG. 16, a pair of auxetic tummy control panels 400 are illustrated on the interior facing side of the front panel 412 of garment 410 on opposite sides of the fly portion 426 of the garment 410. The auxetic tummy control panels 400 are attached beneath the support band 420, e.g., by top stitching 402. The auxetic tummy control panels 400 may be attached along interior side seams 416 of the garment 410. The auxetic tummy control panels 400 may be attached along fly stitching 408 of the garment 410 about the fly portion 426. The auxetic tummy control panels 400 may provide improved tummy support and appearance to the wearer. As shown, the auxetic tummy control panels 400 is free along a bottom edge of each panel; however, in some embodiments, the auxetic tummy control panels 400 may be attached along a bottom edge of the auxetic tummy control panels 400. In certain embodiments, the auxetic tummy control panels 400 may extend above the front panel 412 and may form a portion of the waistband 420 of the garment 410. In such embodiments,

the auxetic tummy control panels **400** may be attached along the top edge of the support band **420**. In particular embodiments, a single auxetic tummy control panel **400** may extend along the inside of the front panel **412** and extend between the side seams **416**.

[0068] FIGS. 17 and 18 illustrate auxetic panels **500** secured to an interior facing side of a front panel **512** of a bottom garment **510** sandwiched between the front panel **512** and a body contact fabric panel **530**. The auxetic panels **500** are similar to the auxetic band **200** with like elements having similar labels with a preceding “5” replacing the preceding “2” of the similar element with only the differences detailed herein for brevity. The auxetic panels **500** are shown in dotted lines with a plurality of unit cells arranged in an array between a first end **502** and a second end **504** of the auxetic panels **500**. The first end **502** and the second end **504** may be attached to the garment **510**. For example, the first end **502** or the second end **504** may be attached to the garment **510** by being sewn into the side seams **516** with the other of the first end **502** and the second end **504** sewn at the fly area **526**. In some embodiments, the attachment is made by stitching **532** and **534** at ends **502** and **504**, respectively. In some embodiments, the auxetic panels **500** may be attached by a third stitching **536** adjacent the top edge of the auxetic panels **500** and adjacent the bottom edge of the support band **520**. The stitching **532**, **534**, and **536** may be separate stitchings or continuous stitchings. In some embodiments, the body contact fabric panel **530** is attached to the garment **510** by the same or similar stitchings **532**, **534**, **536**. In embodiments where there are three stitchings attaching the auxetic panels **500** or the body contact fabric panel **530**, the bottom edge of the sleeve **501** of the auxetic panels **500** is substantially free to move and expand during stretching. When a user’s belly pushes against the auxetic panels **500**, the auxetic panels **500** may expand in all directions increasing surface area and creating comfort during stretch. It is to be understood that either side (right or left) or both sides corresponding to each leg of the bottom garment **510** may incorporate an auxetic panel **500** or other auxetic device.

[0069] The body contact fabric panel **530** provided between the auxetic device **500** and the skin of a wearer may provide a comfort layer in instances where it is preferable to avoid skin contact with the auxetic panels **500**. The body contact fabric panel **530** may be a thin layer of fabric so as to minimize the thickness of layers at that portion of the garment **510**. In some embodiments, the body contact fabric panel **530** is a stretchable fabric.

[0070] In some embodiments, the front panel **512** may include pockets. In such embodiments, the auxetic panels **500** may be placed between the pocket fabric **540** and body contact fabric panel **530**. During construction of the garment **510**, the auxetic panels **500** and the body contact fabric panel **530** may be attached at the same time. For example, the auxetic panels **500** and the body contact fabric panel **530** may be attached as the pocket fabric **540** is installed, allowing for efficiency in creating the garment **510**.

[0071] In certain embodiments, the pocket fabric **540** may be the body contact fabric panel **530** with the auxetic panels **500** disposed between the interior side of the front panel **512** and the pocket fabric **540**. Since pocketing fabric is generally configured to touch a person’s body, the typical pocket fabric already possess desirable comfort characteristics. In

such embodiments, extra processing and material may be minimized as installing pocketing fabric is already a process that the garments undergo.

[0072] Any of the auxetic bands or panels **200**, **300**, **400**, **500** may be provided without a sleeve or cover such that the auxetic bands or panels include an exposed auxetic device, e.g., auxetic device **100**, without cover material or a sleeve. It will be appreciated that as used herein, “auxetic band” may refer to an auxetic band or an auxetic panel. The auxetic devices detailed herein, may be made of an elastomeric material or material that has elastomeric stretch properties. The auxetic devices herein may be manufactured by techniques known in the art including molding or additive manufacturing. In some embodiments, an elastomeric material is provided into a 3D printer and the printer is instructed to print the elastomer material in the shape of an auxetic device, including defining the various unit cell structures that give rise to auxetic properties as detailed below. In some embodiments, a sheet of elastomeric material is provided and the cell structure is imparted to the sheet via a cutting method, including laser cutting and dye cutting. In other embodiments, the elastomeric material is cast into a mold that is configured to provide the various cell structures that give rise to auxetic properties.

[0073] For incorporation into a garment, various factors may be taken into consideration for selection of a material for the auxetic devices detailed herein, e.g., auxetic device **100**. The material of the auxetic devices may have sufficient dimensional stability and may retain its structure after washing, e.g., the material may retain 99 percent or more of its structure. The material of the auxetic devices may be able to survive the various chemicals used to clean fabrics and garments including, but not limited to, detergents, softeners, or bleach. The material of the auxetic devices may not degrade when experiencing the various processes of a wash cycle including exposure to hot water during washing or heat during drying in home washing and/or industrial washing machines. The material of the auxetic devices may exhibit sufficient stretch and recovery while minimizing growth, e.g., less than 10 percent growth.

[0074] In some embodiments, the material of the sleeves or other cover materials, e.g., sleeve **201**, may be a two-way stretch fabric that is a fabric that stretches along one axis and is resistant to stretching in a substantially perpendicular axis. The sleeves or other cover materials may be made of a 2-way stretch fabric formed of stretch cotton blend, stretch cotton/poly blend, cotton/nylon blends, 100 percent poly, 100 percent nylon, or blends and combinations thereof. In some embodiments, the sleeve material is a 2-way stretch denim. The 2-way stretch denim may include a stretch material in the weft and may include a non-stretch material in the warp. Thus, a garment incorporating an auxetic band may appear uniform in material construction. In some embodiments, the sleeves or cover materials may be formed of other two-way stretch materials. In some embodiments, the sleeves or cover materials may be formed of a 4-way stretch material.

[0075] While several embodiments of the disclosure have been shown in the drawings, it is not intended that the disclosure be limited thereto, as it is intended that the disclosure be as broad in scope as the art will allow and that the specification be read likewise. Any combination of the above embodiments is also envisioned and is within the scope of the appended claims. Therefore, the above descrip-

tion should not be construed as limiting, but merely as exemplifications of particular embodiments. Those skilled in the art will envision other modifications within the scope of the claims appended hereto.

What is claimed:

1. A garment comprising:  
an auxetic device having a first end, a second end opposite the first end, a top edge extending between the first end and the second end, and a bottom edge opposite the top edge, the first end and the second end being fixed to another portion of garment, at least one of the top edge or the bottom edge being free between the first end and the second end, the auxetic device having a Poisson ratio less than or equal to zero when stretched in a direction extending between the first end and the second end.
2. The garment according to claim 1, wherein the auxetic device includes a plurality of unit cells disposed between the first end and the second end.
3. The garment according to claim 2, wherein the plurality of unit cells are arranged to have a reentrant structure.
4. The garment according to claim 1, further comprising:  
a front panel and a rear panel joined together at side seams to form outer edges of the garment, the front panel and the rear panel define a top opening configured to be positioned adjacent a waist of a wearer; and  
a support band extending about the top opening, the support band configured to maintain a position of the garment relative to the waist of a wearer.
5. The garment according to claim 4, wherein the auxetic device forms a portion of the support band, the auxetic device forming a portion of the support band above the rear panel or forming a portion of the support band above one of the side seams.
6. The garment according to claim 4, wherein the first end is attached to the support band adjacent one of the side seams defined between the front panel and the rear panel and the second end is attached to the support band adjacent the other side seam.
7. The garment according to claim 4, wherein the front panel includes a fly assembly having a first side and a second side, the fly assembly configured to selectively secure one side of the front panel to another side of the front panel, the first end of the auxetic device attached to the first side of the fly assembly and the second end attached to the second side of the fly assembly.
8. The garment according to claim 1, further comprising a sleeve that defines a tunnel, the first end and the second end of the auxetic device each including a solid area portion, the sleeve closed about each solid area portion of the first end and the second end such that the auxetic device is enclosed within the sleeve.
9. The garment according to claim 1, further comprising:  
a front panel and a rear panel joined together at side seams to form outer edges of the garment, the front panel and the rear panel define a top opening configured to be positioned adjacent a waist of a wearer; and  
a body contact fabric, the auxetic device sandwiched between the front panel and the body contact fabric, the body contact fabric having a first edge and a second edge opposite the first edge, the auxetic device attached to the front panel or the rear panel along one of the first edge or the second edge, the body contact fabric configured to conceal the auxetic device.

10. The garment according to claim 9, wherein the first end and the first edge are stitched together to attach the auxetic device and the body contact fabric to the front panel or the rear panel.

11. An auxetic band for a garment, the auxetic band comprising:

- an auxetic device having a first end, a second end opposite the first end, a top edge extending between the first end and the second end, and a bottom edge opposite the top edge, the auxetic device having a Poisson ratio less than or equal to zero; and
- a sleeve of stretch fabric having a first sleeve end and a second sleeve end opposite the first sleeve end, the sleeve of stretch fabric dimensioned to house the auxetic device therein, the first end of the auxetic device fixed to the sleeve adjacent the first sleeve end and the second end of the auxetic device fixed to the sleeve adjacent the second sleeve end, at least one of the top edge or the bottom edge free to move relative to the sleeve.

12. The auxetic band according to claim 11, wherein the sleeve is dimensioned to accommodate an increase in surface area when the auxetic device is stretched in a direction extending between the first end and the second end.

13. A garment comprising:

- an auxetic band according to claim 11, wherein a top edge of the auxetic band is positioned at or below a waistband of the garment or a bottom edge of the auxetic band is positioned at a waistband of the garment.

14. A garment comprising:

- a support band configured to position a top portion of the garment about or adjacent a waist of a wearer;
- an auxetic device having a first end and a second end opposite the first end, the auxetic device having a Poisson ratio less than or equal to zero, the auxetic device secured to the support band such that at least one edge of the auxetic device is free to move with respect to the support band.

15. The garment according to claim 14, wherein the auxetic device forms a portion of the support band.

16. The garment according to claim 14, wherein the auxetic device includes a top edge secured to the support band or below the support band such that the auxetic device forms a belly support panel of the garment below the support band.

17. The garment according to claim 16, wherein the auxetic device includes a bottom edge, the bottom edge substantially free from the rest of the garment.

18. The garment according to claim 14, wherein the auxetic device extends above the support band of the garment.

19. The garment according to claim 14, further comprising:

- a front panel; and
- a body contact fabric panel secured to the front panel, the auxetic device positioned between the body contact fabric panel and the front panel with the body contact fabric panel preventing the auxetic device from contacting the skin of a wearer.

20. The garment according to claim 19, further comprising a pocket formed of pocket fabric, the pocket fabric disposed between the front panel and the auxetic device.